

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving StudentsPerformance.csv to StudentsPerformance (1).csv

```
df = pd.read_csv('StudentsPerformance.csv')
df.head()
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	read score
0	female	group D	some college	free/reduced	none	69	
1	male	group C	some college	free/reduced	completed	100	
2	female	group C	some high school	standard	completed	44	
3	female	group D	high school	free/reduced	none	62	
4	female	group D	some college	standard	completed	84	

```
df.shape
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   gender                                1000 non-null   object
1   race/ethnicity                        1000 non-null   object
2   parental level of education           1000 non-null   object
3   lunch                                 1000 non-null   object
4   test preparation course               1000 non-null   object
5   math score                            1000 non-null   int64
6   reading score                         1000 non-null   int64
7   writing score                          1000 non-null   int64
dtypes: int64(3), object(5)
memory usage: 62.6+ KB
```

	0
gender	0
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	0
reading score	0
writing score	0

dtype: int64

```
df['average_score'] = (df['math score'] + df['reading score'] + df['writing score'])
```

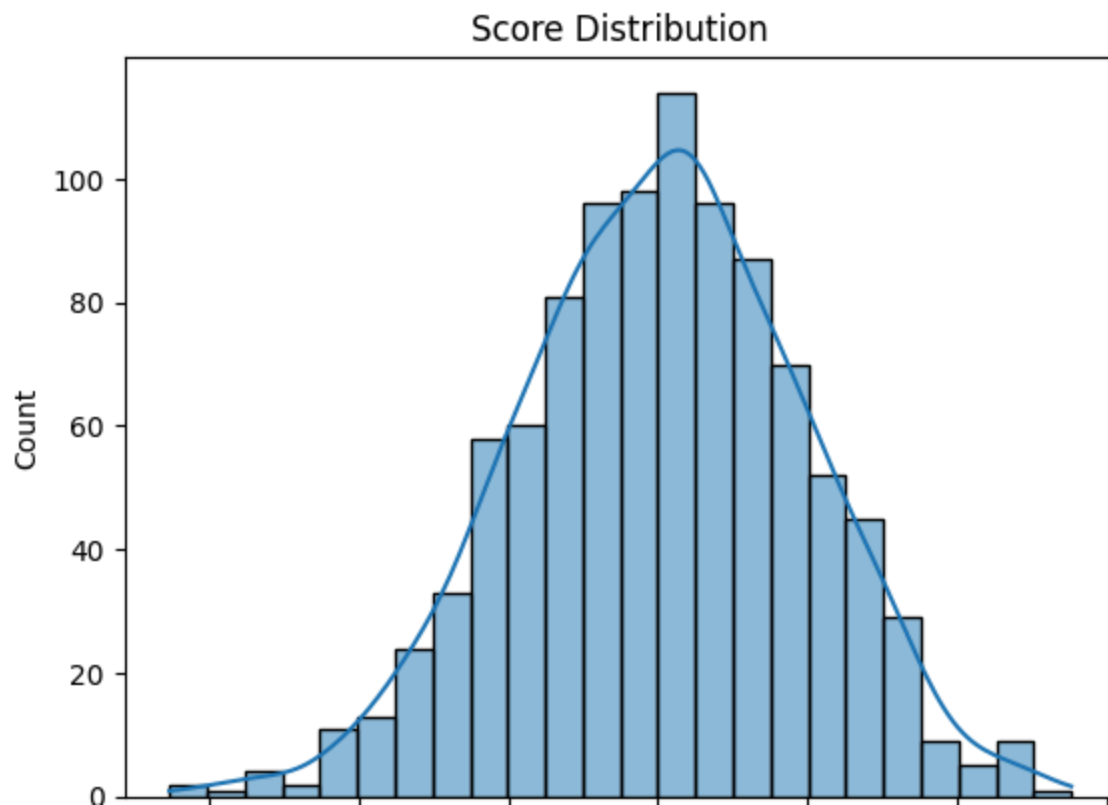
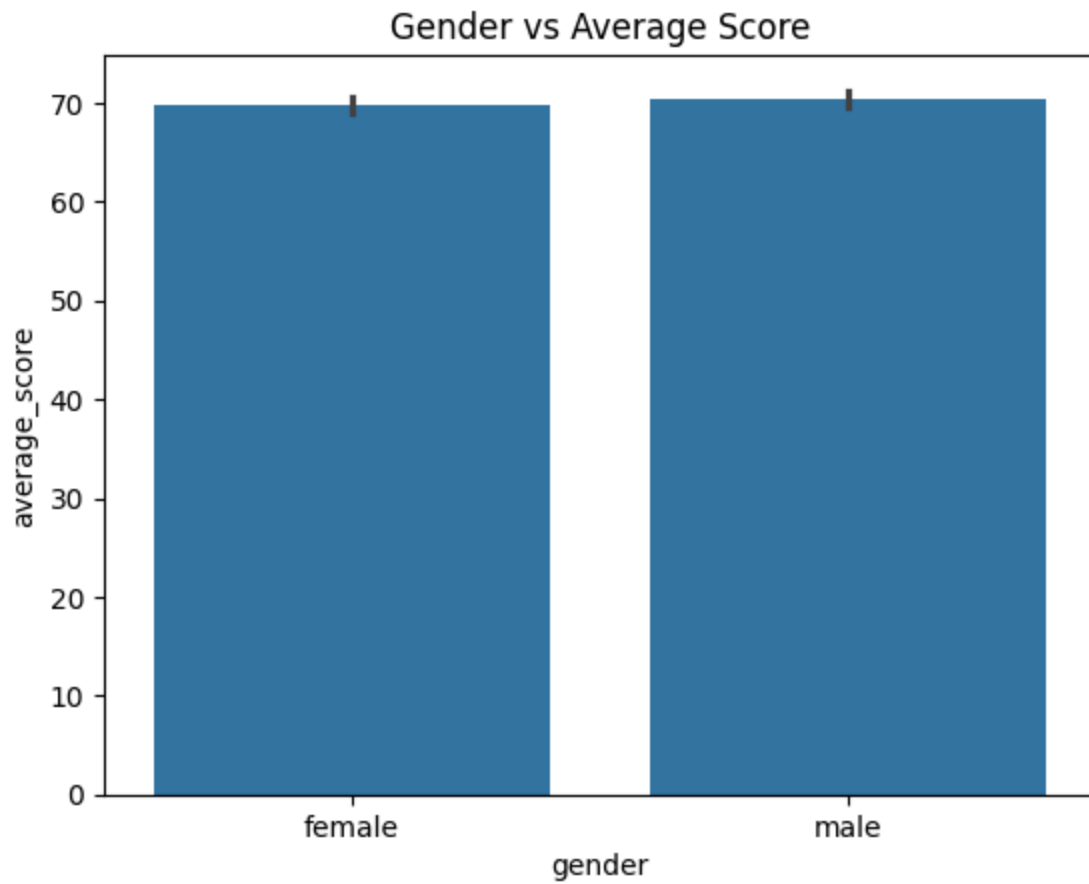
```
# Gender vs Average Score
sns.barplot(x='gender', y='average_score', data=df)
plt.title('Gender vs Average Score')
plt.show()

# Score Distribution
sns.histplot(df['average_score'], kde=True)
plt.title('Score Distribution')
plt.show()

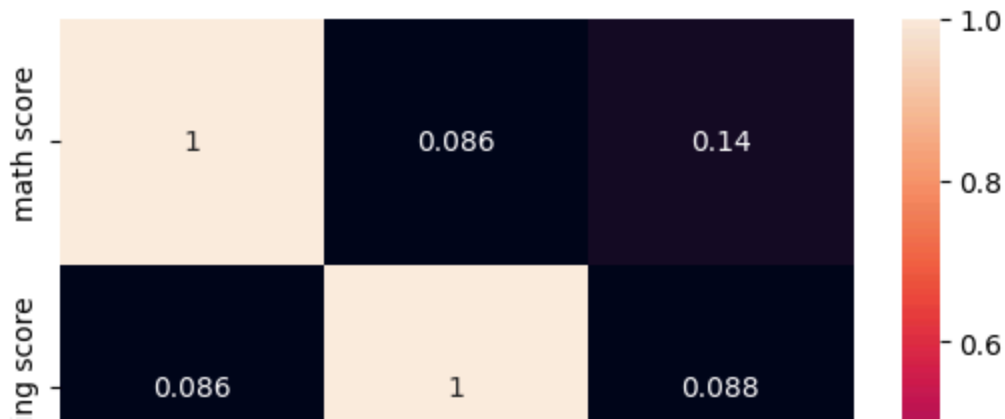
# Test Preparation vs Score
sns.barplot(x='test preparation course', y='average_score', data=df)
plt.title('Test Prep vs Score')
plt.show()

# Parental Education vs Score
sns.barplot(x='parental level of education', y='average_score', data=df)
```

```
plt.xticks(rotation=45)
plt.title('Parental Education vs Score')
plt.show()
```



```
sns.heatmap(df[['math score','reading score','writing score']].corr(), annot=True,
plt.show()
```



```
df['result'] = df['average_score'].apply(lambda x: 'Pass' if x>=40 else 'Fail')

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression

le = LabelEncoder()
df['test_prep_encoded'] = le.fit_transform(df['test preparation course'])

X = df[['test_prep_encoded']]
y = df['result']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LogisticRegression()
model.fit(X_train, y_train)
print("Accuracy:", model.score(X_test, y_test))
```

Accuracy: 0.995

```
df.to_csv('final_student_data.csv', index=False)
from google.colab import files
import
```

```
sns.barplot(x='test preparation course', y='average_score', data=df)
plt.title('Test Prep vs Score')
plt.savefig('test_prep_vs_score.png')    # ముందు save cheyandi
plt.show()                               # తర్వాత show cheyandi
```

